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Connecting Teacher Professional Development and Student Mathematics Achievement: A 4-Year Study of an Elementary Mathematics Specialist Program

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Abstract

The present study investigated the effects of *Primarily Math*, an inservice elementary mathematics specialist program. *Primarily Math* sought to augment the mathematical knowledge for teaching of kindergarten through third-grade teachers using a longitudinal multiple cohort design. Two sets of analyses were conducted. The first examined impact on teachers' mathematical knowledge for teaching, attitudes toward learning mathematics, and beliefs about teaching and learning relative to a matched comparison group. *Primarily Math* teachers demonstrated greater knowledge for teaching Numbers and Operations and more positive attitudes toward learning mathematics, and more often endorsed student-centered beliefs about teaching and learning. The second set of analyses examined the extent to which students of three cohorts of *Primarily Math* teachers demonstrated more fall–spring growth in a measure of mathematics achievement relative to students of comparison-group teachers. There was a small but positive effect of participation in *Primarily Math* on student mathematics achievement.

Keywords

professional development, mathematics education, quantitative research, elementary education

Teachers and teaching are at the center of mathematics education reform (National Research Council, 2011). The prevailing assumption that teachers learn what they need to know before they enter the classroom has been dispelled (Elmore, 2002) and professional development (PD) has been recognized as an avenue to school improvement. Ginsburg, Lee, and Boyd (2008) identified two critical issues influencing the quality of mathematics instruction and achievement in the early years, which have become clear targets of PD efforts: (a) teachers' poor mathematical preparation, and (b) teachers' fear of mathematics and desire to avoid mathematics. Yet, the field does not have a complete understanding of what combinations of professional learning activities are most appropriate and effective for early primary teachers.

The National Science Foundation created the Math–Science Partnership program with the goal of improving student achievement in mathematics and science by developing and delivering PD through partnerships among K-12 districts and institutions of higher education. *Primarily Math* is one such PD opportunity for kindergarten through third-grade teachers. This study examines how *Primarily Math* impacted (a) multiple cohorts of teachers' knowledge for teaching mathematics, their attitudes toward learning mathematics, and their beliefs about teaching and learning; and (2) observed growth in students' mathematics ability over an academic year. In addition, we sought to determine what teacher-level professional resources and characteristics were associated with growth in students' mathematics achievement.

What Do We Think High-Quality PD in Early Mathematics Looks Like?

Wilson and Berne (1999) suggested high-quality PD affords teachers opportunities to talk about subject matter, students,

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learning, and teaching. Borko, Jacobs, and Koellner (2010) outlined the content and process/structural characteristics of effective PD, and Darling-Hammond and Richardson (2009) compiled a list of PD practices supported by research. These PD practices align with the work of Desimone and colleagues (e.g., Birman, Desimone, Porter, & Garet, 2000; Desimone, 2009; Garet, Porter, Desimone, Birman, & Yoon, 2001) who have written extensively on five proposed components of effective PD (see Desimone, 2009, for a review).

First, teachers need opportunities to deepen their *content knowledge*. Second, teachers need to engage in *active learning*, with opportunities to observe peers or master teachers, receive feedback and discuss their pedagogical judgment, and analyze student work. Third, PD activities need to be *coherent*: Learning must resonate with (but not necessarily mirror) teacher knowledge and beliefs, as well as align with the policies and culture of the teaching context they will return to once PD ends. Fourth, PD experiences must be of sufficient *duration*: Garet et al. (2008) suggested activities be spread over the semester, including at least 20 hr of contact time with the instructor or PD provider. Fifth, PD activities should cultivate *collective participation* among teachers, building communities of discourse. Research on these five practices in ongoing and Wayne, Yoon, Zhu, Cronen, and Garet (2008) importantly reminds us that to date, there is insufficient detail to guide best practices.

The Targets of Past PD Programs in Elementary Mathematics

Understanding the effect of teachers and teaching likely requires researchers to look beyond traditional teacher characteristics (e.g., certification; Phillips, 2010) and instead focus on more direct indicators of teacher knowledge, attitudes, and beliefs as key factors influencing student achievement. Contemporary ideas of effective PD include activities designed to nurture and replenish (a) robust mathematical knowledge for teaching, (b) positive attitudes toward learning mathematics, and (c) constructivist beliefs about teaching and student learning—all professional resources that leverage teachers' capacity for high-quality instruction (Borko et al., 2010; Stein, Smith, & Silver, 1999). These skills and capacities were targets of *Primarily Math* and are the focus of this study.

Mathematical knowledge for teaching. The knowledge and skills necessary to do mathematics are different from those required to teach mathematics, which include, for example, posing mathematical questions, giving and appraising explanations, and selecting and using representations (Thames & Ball, 2010). Knowledge for teaching mathematics enables teachers to analyze and understand student thinking and provide developmentally appropriate support and strategies, while maintaining the integrity of the content (Hill, Schilling, & Ball, 2004; Learning Mathematics for Teaching Project [LMT], 2011). Hill et al. (2004) described their theory of mathematical knowledge for teaching, and empirically validated its multidimensionality. Hill, Ball, Blunk, Goffney, and Rowan (2007) as well as Hill, Blunk, et al. (2008) demonstrated specialized content knowledge was correlated with higher quality mathematics instruction. Finally, Hill, Rowan, and Ball (2005) demonstrated mathematical knowledge for teaching predicted the size of student gain scores, even after controlling for the effect of student-level characteristics (e.g., socioeconomic status [SES]) as well as teacher-level characteristics (e.g., teaching credentials).

Attitudes toward learning. Teachers' emotional stances toward an academic domain are consequential: Teacher attitudes can impact student learning and skills acquisition within that domain. Mathematics anxiety is a prevalent condition within the population of early childhood educators (Ginsburg et al., 2008; Hembree, 1990). Gunderson, Ramirez, Beilock, and Levine (2013) found first- and second-grade teachers' spatial anxiety predicted students' spatial skills at the end of the year. Beilock, Gunderson, Ramirez, and Levine (2010) also provide evidence suggesting anxiety may be "contagious." At the beginning of the school year, there was a nonsignificant correlation between first- and second-grade female teachers' mathematics anxiety and student achievement. However, by the end of the academic year, gender ability beliefs mediated the relationship between teachers' levels of mathematics anxiety and student achievement; the more anxious female teachers were toward mathematics, the more likely girls (but not boys) were to endorse the stereotype "girls are good at reading and boys are good at mathematics." Moreover, these girls also demonstrated lower levels of mathematics achievement.

Beliefs about teaching and learning. Beliefs about teaching and student learning underpin teacher choices, judgment, and behavior in the classroom (Pajares, 1992; Pintrich, 1990; Staub & Stern, 2002; Voss, Kleickmann, Kunter, & Hachfeld, 2011). Raymond (1997) identified past learning experiences as primary sources of mathematics-related beliefs for elementary teachers, and Pajares (1992) suggested beliefs constructed early in life had the power to persevere even against contradictions caused by reason, time, schooling, and experience. However, PD programs focused on student thinking have provided evidence that teacher beliefs can be modified. For example, Schifter and Fosnot (1993) reported teachers using *Young Mathematicians at Work* concurrently changed their beliefs and instructional practices to reflect more constructivist principles. Fennema et al. (1996) reported that 18 out of 21 teachers who participated in *Cognitively Guided Instruction* over the course of 4 years evolved from demonstrating procedures to creating mathematically richer problem-solving situations. These 18 teachers also encouraged students to share their mathematical thinking, reflecting student-centered beliefs (Capraro, 2001). Although teachers' beliefs are relatively stable psychological attributes, participation in targeted PD can alter beliefs.

PD Program Impacts

To date, there is evidence that university-based partnerships (e.g., *Cognitively Guided Instruction*: Carpenter, Fennema, Peterson, Chiang, & Loef, 1989; *MyTeachingPartner-Math*: Kinzie et al., 2014; McGuire & Kinzie, 2013) and state efforts investing in PD (e.g., Kentucky: Borko, Elliott, & Uchiyama, 2002; Title I schools in seven states: Desimone, Smith, & Phillips, 2013) have positive impacts on early mathematics achievement. Program content and structure reflect ideas about what effective PD requires. However, the preponderance of evidence supporting the link between these practices to desired outcomes is limited (Yoon, Duncan, Lee, Scarloss, & Shapley, 2007) within mathematics education as well as other subject domains (e.g., Penuel, Fishman, Yamaguchi, & Gallagher, 2007, in science; Garet et al., 2008, in literacy).

Desimone (2009) outlined a framework for how to study the impact of PD on teachers and students, which begins with testing teacher change. Nonetheless, it is challenging to demonstrate how PD components (individually and cumulatively) are effective, especially when the standard of evidence is statistically significant gains in student achievement. Indeed, Knapp (2003) stated, " . . . the actual benefits of the asserted attributes of good PD have rarely been examined systematically or in combination" (p. 120); more than 10 years later, this statement still reflects the state of the research.

The purpose of the current study is to contribute to our understanding of how intensive content-focused PD impacts teacher and student outcomes. More specifically, we study the extent to which the combination of *Primarily Math* PD learning opportunities—characterized by *content*, *active learning*, *coherence*, *duration*, and *collective participation*—change teacher knowledge, attitudes, and beliefs, as well as student growth in an assessment of early mathematics ability.

Primarily Math Program

Theory of Teacher Change

The architects of the *Primarily Math* have been providing PD for more than 20 years. Several of the original grant's principal investigators are included as senior authors for this article, representing expertise in mathematics, mathematics education, and early childhood education, as well as in designing and delivering PD for teachers. The lead author and graduate research assistants who helped to collect and analyze data, and sometimes helped deliver the PD, were not the original architects of *Primarily Math*, and as such were able to provide unbiased analyses. Data analyses cannot be interpreted separate from the related contexts; therefore, it is necessary for data analyses to be conducted by those familiar with the design, nature, and structure of the *Primarily Math* program. The *Primarily Math* research team had regular discussions throughout the process of design, data collection, and data analyses to check for biases, to search for disconfirming evidence, and to ensure conclusions were supported by evidence in the datasets. Thus, the *Primarily Math* framework of how teachers learn and change in the context of PD is grounded in wisdom of practice as well as theory and research. We assume, first, teachers have powerful (but often implicit and unarticulated) preexisting images of what it means to be an effective teacher, coupled with ideas about how students learn and should behave (Lortie, 1975; Pajares, 1992). Thus, change may be slow and difficult (Guskey, 2002). Second, teachers change their beliefs about teaching in related but nonlinear ways (Fullan, 2001). Some teachers may need to change their beliefs before they alter their teaching while other teachers may need to make changes to their practice before their beliefs change. Third, change is provoked through inquiry and ongoing reflection *in* and *from* practice (Cochran-Smith & Lytle, 2009; Lampert, 2001).

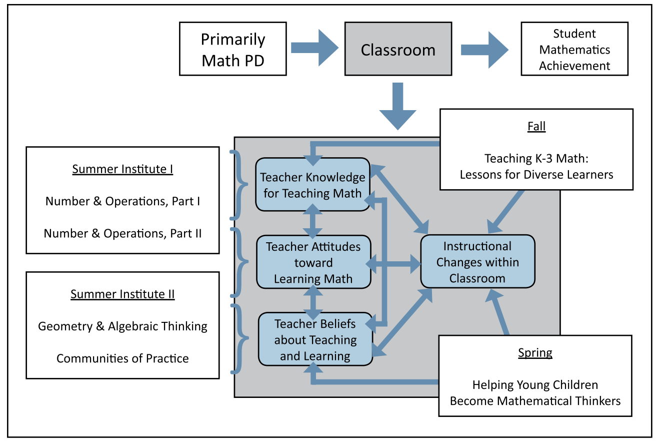

Figure 1. *Primarily Math* logic model.

Figure 1 depicts *Primarily Math's* logic model. PD inputs consisted of six graduate-level courses taken over 13 months.[1] The first Summer Institute was composed of two mathematics courses. These were followed by two pedagogy courses during the academic year, in which teachers were given assignments that took advantage of their access to students. The second Summer Institute included one final mathematics course and one pedagogy course.

Within the classroom (delineated by the gray box), teacher knowledge for teaching mathematics, their attitudes toward their own professional learning, and their beliefs about teaching and learning are interrelated psychological attributes, as depicted by bidirectional arrows. Teacher knowledge, attitudes, and beliefs serve as a filter through which teachers make sense of their PD experiences (including coursework); the coursework in turn is intended to collectively impact teacher knowledge, attitudes, and beliefs (as depicted by bidirectional arrows). Teacher knowledge, attitudes, and beliefs (bidirectionally) as well as pedagogy coursework (unidirectionally), in turn, influence instruction within the classroom. Overall, we expect opportunities to learn from summer institutes, and teachers' own practice represents a collection of experiences that are associated with changes in instruction and will have positive impacts on student achievement.

Importantly, opportunities to learn in *Primarily Math* align with the five proposed core components of effective PD (as defined above through the work of Desimone and colleagues). The appendix contains the core components, a description of how the *Primarily Math* course content and structure aligns with each component, and examples.

Research Questions

Primarily Math's combination of learning opportunities, settings, and partners may foster multidimensional change in how teachers think through and make pedagogical decisions. Thus, this study addresses three research questions.

> **Research Question 1:** What is the impact of *Primarily Math* on teachers' mathematical knowledge for teaching, their attitudes toward learning mathematics, and their beliefs about teaching and learning relative to a matched comparison group?
> **Research Question 2:** To what extent do the students in the classrooms of *Primarily Math* teachers differ in their mathematics achievement relative to the students of teachers in a comparison group, as measured by growth in math ability scores from fall to spring within an academic year?
> **Research Question 3:** To what extent will other variables—such as SES (measured at the school building level) and time spent in the study (expressed as the measurement occasion for a teacher)—combine with measures of teacher knowledge, attitudes, and beliefs about mathematics teaching and learning to have an impact on student math achievement?

Method

The *Primarily Math* research agenda focused on three cohorts of teachers and a matched comparison group. Starting dates of each cohort were staggered: Cohort 1 began in summer 2009, followed by Cohort 2 in summer 2010, and Cohort 3 in summer 2011.

At the request of district partners, cohorts were constructed based on the geographic location of participating teachers. District partners hoped *Primarily Math* teachers would emerge as leaders who would share what they learned with teachers in their building and, thus, preferred teachers from the same building and district participate in *Primarily Math* during the same year.

All three cohorts were recruited in 2009. Teachers applied to participate in *Primarily Math* by completing an application, which was reviewed by university and school district personnel. Selection criteria were based on evidence of positive teaching practices, leadership, and attitude toward participation (motivation to learn); teachers in high-need schools (based on percentage of students receiving free/reduced price lunches) were given preference. The *Primarily Math* partnership team developed a scoring rubric for participant applications after collectively agreeing on the factors that would comprise the score. Higher scores were awarded to teachers who presented evidence of lifelong learning (past PD participation), the desire to improve their mathematics teaching practices, and strong written communication skills.

Overall, 96% of teachers who began *Primarily Math* coursework also completed the 18-credit graduate-level program of study.[2] After passing the six courses, teachers received a K-3 Mathematics Specialist certificate from the university.

Sample

Primarily Math *teachers and students.* *Primarily Math* participants came from across the state. For this study, however, all analyzed teacher and student data came from three large, urban school districts. Student data were only collected in a subset of *Primarily Math* and comparison-group teacher classrooms. We focus on this subset to link student growth in mathematics achievement to individual teachers. We followed the teachers over time, so students in each classroom changed from one academic year to the next. Table 1 contains the number of *Primarily Math* teachers for each grade level per cohort by district. Table 2 contains the baseline demographics of the analyzed (subset) sample of *Primarily Math* teachers and the comparison group. District A had more participants than Districts B or C, partly due to district size and partly due to the relative success of recruitment efforts. District C contributed the most teachers in the comparison group as part of the district's desire to better understand what their teachers knew and could do related to mathematics teaching and learning.

Although K-3 teachers participated in the PD program, we found about half of third graders scored too high in the fall to detect growth in mathematics achievement—an unanticipated limitation of our selected assessment. Thus, we restricted the research study data to K-2 classrooms. Table 2 shows *Primarily Math* and comparison-group participants from the three target districts whose classrooms contributed child data to the study; Table 1 shows *Primarily Math* and comparison teachers overall, whether their students were tested or not.

The individual teacher was the recipient of PD and therefore the unit of replication; students within classrooms were the unit of analysis. A preliminary power analysis demonstrated the need to test at least 10 students per classroom to detect programmatic effects at an alpha level of .10 instead of the customary .05.[3] This standard was generally achieved across *Primarily Math* and comparison teachers. Within each classroom, each fall, 12 to 15 students (to account for student mobility) were randomly selected from consenting students and families to participate in the mathematics assessment. Policies of district partners did not

Table 1. Teachers by Grade-Level, Cohort, District Whose Students Were Tested.

District	Grade level	Cohort 1	Cohort 2	Cohort 3	Comparison group
District A	Kindergarten	4	0	19	12
	First grade	7	1	8	4
	Second grade	4	0	5	7
	Third grade	2	0	5	7
District B	Kindergarten	0	4	0	2
	First grade	0	1	0	0
	Second grade	0	7	0	5
	Third grade	0	7	0	2
District C	Kindergarten	0	0	0	20
	First grade	0	2	0	20
	Second grade	0	0	0	20
	Third grade	2	2	0	5

Table 2. Teacher Characteristics of *Primarily Math* and Comparison Teachers.

Teacher characteristics	*Primarily Math* (Cohorts 1-3) (n = 126)		Comparison group (n = 92)	
Mean years of teaching experience	9.97 years		9.83 years	
Highest degree attained: BA or BS	49%		44%	
Highest degree attained: MA, MS, or MEd	47%		50%	
Grade level taught	K	25%	K	31%
	First	14%	First	24%
	Second	24%	Second	28%
	Third	19%	Third	7%
	Other	18%	Other	9%

Note. Teachers in the "Other" category include those teaching multiple grade levels, grade levels other than K-3, special education, English language learners, and mathematics coaches.

permit access to individual student-level data such as free/reduced lunch or English language learner (ELL) status. Instead, we used publicly available building-level data such as the free/reduced lunch percentage for the school.

Comparison group. A comparison group was recruited from buildings in partner districts with no *Primarily Math* participants. Teachers were matched based on three building-level characteristics: the proportion of students who qualified for free/reduced price lunch (which we consider a proxy for SES), the total number of students, and the racial/ethnic composition of students.

Data collection began in 2009. Table 3 depicts measurement occasions for each cohort and the comparison group. The "pretest" is the measurement occasion prior to the start of *Primarily Math* coursework. The "posttest" is the measurement occasion after completion of the sequence of *Primarily Math* coursework. Finally, "follow-up" is the measurement occasion where teachers completed assessment and survey materials during the summer after exiting the *Primarily Math* program. In addition, as shown in Table 3, Cohorts 2 and 3 have "baseline" data. We define "baseline" as all the measurement occasions prior to the "pretest." The same kind of teacher-level and student-level data were collected for baseline occasions, so that the research team could monitor the fluctuation of teachers' and students' outcomes before teacher participation in *Primarily Math*.

Instruments

Teacher-level data were collected annually in the spring with two exceptions. First, Cohort 1's posttest started later in fall 2010 relative to the timing of Cohorts 2 and 3. Second, Cohort 2 and 3 posttests were administered prior to coursework completion; specifically, teachers had yet to complete the last two summer courses. Student achievement data were collected in fall and spring from 2009 to 2013. Data collection in the fall usually spanned September and October, while data collection in the spring usually spanned March and April. The only exception to this pattern was the first year of data collection (fall 2009), when logistical procedures and protocols were being piloted and testing spanned September through mid-November.

Mathematical Knowledge for Teaching survey (MKT). The MKT (Hill et al., 2004) was used to assess teachers' mathematical knowledge unique to teaching. This instrument is composed of 36 and 34 questions for Versions A and B, respectively. There are multiple parts per item within three subscales: Numbers and Operations; Patterns, Functions, and Algebra;

Table 3. Measurement Occasions for Cohorts 1 to 3 and Comparison-Group Teachers (2009-2012).

	2009	2010	2011	2012
Cohort 1	Pretest	Posttest	First year follow-up	Second year follow-up
Cohort 2	Baseline 1	Pretest	Posttest	First year follow-up
Cohort 3	Baseline 1	Baseline 2	Pretest	Posttest
Comparison group	Measurement occasion	Measurement occasion	Measurement occasion	Measurement occasion

and Geometry. Raw scores are transformed onto an item response theory (IRT) scale: A 0 indicates a score falls at the national norm for a K-6 teacher. Teachers alternated between the two versions to reduce potential retest bias, and they completed Version A in 2009 and 2011, and Version B in 2010 and 2012.

Fennema–Sherman Mathematics Attitudes Scales for Teachers (FSMAS). When planning *Primarily Math* in 2008, the research team was unable to locate any measure of teachers' attitudes toward their own learning of mathematics that had been widely and consistently used in existing research. Thus, the research team selected and adapted three scales from the FSMAS (Fennema & Sherman, 1976; Ren, Green, & Smith, 2016) originally developed to assess students' attitudes toward learning mathematics: (a) the Confidence in Learning Mathematics Scale (10 items); (b) the Mathematics Anxiety Scale (12 items); and (c) Effectance-Motivation Scale in Mathematics (11 items). The Confidence scale measures one's confidence in his or her abilities to learn and perform well in mathematics; the Anxiety scale assesses individuals' feelings of anxiety, nervousness, and dread related to doing mathematics; and the Effectance-Motivation scale measures whether one enjoys and seeks challenges regarding mathematics. The adaptation and validation procedures have been reported in detail in Ren et al. (2016). Responses were measured using a 5-point rating scale, from 1 (*strongly disagree*) to 5 (*strongly agree*). Coefficient alpha for these three subscales are as follows: Confidence, $\alpha = .92$; Anxiety, $\alpha = .94$; and Effectance-Motivation, $\alpha = .93$.

Mathematics Beliefs Scales (MBS). The MBS (Capraro, 2001; Fennema, Carpenter, & Loef, 1990; Ren & Smith, 2013) were designed to measure teachers' beliefs about teaching mathematics and how students learn mathematics using a 5-point rating, from 1 (*strongly disagree*) to 5 (*strongly agree*). For this project, we used the three subscales from Capraro's (2001) 18-item short-form: Student Learning, Stages of Learning, and Teacher Practices. Confirmatory factor analyses using baseline data from *Primarily Math* teachers indicated Capraro's (2001) factor structure did not fit our data. Instead, exploratory factor analyses indicated a two-factor solution, which we called Student-Centered Beliefs and Teacher-Centered Beliefs. Teachers who embrace student-centered teaching believe students construct their own knowledge through active investigation and meaningful exploration. Teachers who value teacher-centered teaching believe students need to be told or shown how to do mathematics and value efficient problem-solving paths that lead to correct answers. Coefficient alphas for the two subscales are as follows: Student-Centered Beliefs, $\alpha = .78$; and Teacher-Centered Beliefs, $\alpha = .86$.[4]

Test of Early Mathematics Ability–Edition 3 (TEMA-3). The TEMA-3 (Ginsburg & Baroody, 2003) assesses the formal and informal mathematical (number and operations) knowledge of children aged 3 years to 8 years 11 months. Raw scores are converted to math ability scores that provide an indication of a child's overall mathematics achievement. A math ability score of 100 represents the 50th percentile and has a standard deviation of 15. A child who has a math ability score between 90 and 110 is considered to be "Average." Math ability scores depend both on raw score and on age, so a "typically" achieving student would score 100 at each testing occasion. Thus, a growth score of 0 indicates typical growth.

The test administrators were graduate students, retired primary teachers, and community members. They attended a 2-hr training session and practiced administering the TEMA-3 to at least three children while under observation of the trainer. Test administrators noted any disruptions that occurred during testing (e.g., student needed to use the bathroom, fire drills) as well as any other factors that could affect performance (e.g., teacher reported the child was having a bad day). If either the test administrator or trainer judged data to be compromised by a disruption, they were excluded from the analysis.

An additional note about the selection of the TEMA-3 is the relative dearth of validated mathematics assessments for young primary students. Initially, the research team was planning to use the math sections of the Peabody Individual Achievement Test (PIAT); however, the PIAT is general in nature, while the TEMA-3 focuses on number and operations, which is the same focus as the *Primarily Math* coursework. While the small pool of available instruments could be considered a limitation, the TEMA-3 is a good match for the *Primarily Math* focus, and thus was determined to be the best instrument to use.

Data Analysis

Research Question 1, which addresses the impact of *Primarily Math* at the teacher level, was examined using a series of growth models of change in teacher outcomes over time for *Primarily Math* (treatment) and the comparison group. This is a widely accepted modeling approach for studying change over time within and between individuals. A separate model was fit to each of the teacher outcomes (i.e., math content knowledge in the three separate topic areas, confidence, anxiety, effectance/motivation, teacher-centered beliefs, and student-centered beliefs). The independent variables included participation in *Primarily Math* (a dichotomous indicator with 0 representing the comparison group and 1 representing *Primarily Math*), time in study (a continuous variable representing number of years participating in the study), and the interaction between them. We specified a random effect for teacher intercepts, which allowed each teacher to have a unique starting estimate for the given teacher outcome to account for baseline differences in individual scores.[5]

Research Questions 2 and 3 were examined within a Hierarchical Linear Modeling (HLM) framework. We define Math Ability Score Difference (MASD) as the difference in a student's age-adjusted score on the TEMA-3 assessment. Thus, a score of 0 indicates the child's increase in math achievement is consistent with what is expected for the child. A positive MASD score indicates the student gained more than expected, and a negative MASD score indicates the student gained less than expected. The change score MASD served as the dependent variable in a series of models.

The following variables were examined as predictors of math achievement change: SES; participation in *Primarily Math*; fall math ability score, which varied by student, time point, and teacher; and measurement occasion (a continuous variable representing a teacher's measurement occasion within the study ranging from 0 to 3). The interaction between treatment and time was also tested, along with a random effect for time at the teacher level. The effect for time was estimated as a fixed effect, rather than by including a separate level of nesting for time point. Fall math ability scores and SES were each grand-mean-centered, and time was recoded so the first time point was zero. The treatment variable was coded so a positive effect would represent a higher mean MASD for the treatment group than the comparison group. Three sets of teacher measures, treated as outcomes for the first research question, were also examined, this time as predictors of growth in MASD.

A base model was first fit to the data, containing only random intercepts for teachers. Variables were then added sequentially, and statistically tested based on their contribution to model fit compared with the previous, less complex model. Improvement in model fit was determined using likelihood ratio chi-square tests (χ^2), Akaike information criterion (AIC), and Bayesian information criterion (BIC). If a variable improved model fit, it was retained in the model; otherwise, it was removed before subsequent variables were added and tested.[6]

Results

The Impact of Primarily Math *on Teachers*

Table 4 contains the fixed effect estimates, standard errors, *t* values, and corresponding *p* values for the growth models for each teacher outcome (Research Question 1). With time in study coded as 0, 1, and 2, intercepts are interpreted as the mean outcome for comparison-group teachers during their first year in the study. The treatment effect then estimates the mean increase over the intercept for *Primarily Math* teachers. The time slope is the mean increase in the outcome for the comparison group for an increase of 1 year in the study, and the interaction between treatment and time is the additive effect on this time slope for *Primarily Math* teachers. Thus, for the Number & Operations outcome, comparison teachers were estimated to have an intercept in the IRT metric of −0.47, and the intercept for *Primarily Math* teachers was −0.47 + 0.53 = 0.06. The time slope for comparison teachers was −0.07, and for *Primarily Math* teachers was −0.07 + 0.31 = 0.24. Thus, treatment teachers were estimated to start with a higher numbers IRT score, and to increase at 0.24 points per year in *Primarily Math*.

Primarily Math teachers also started with statistically higher knowledge for teaching Geometry as well as Patterns, Functions,

Table 4. Fixed Effects From Teacher Outcome Growth Models.

Outcome	Effect	Estimate	*SE*	*t* value	*p* value
Numbers and Operations	Intercept	−0.47	0.15	−3.03	.002
	Treatment	0.53*	0.22	2.35	.020
	Time	−0.07	0.09	−0.77	.443
	Treatment:Time	0.31*	0.14	2.28	.025
Geometry	Intercept	−0.44	0.14	−3.18	.002
	Treatment	0.45*	0.21	2.18	.031
	Time	0.07	0.08	0.92	.361
	Treatment:Time	0.15	0.12	1.21	.223
Patterns, Functions, and Algebra	Intercept	−0.75	0.15	−4.94	<.001
	Treatment	0.83*	0.24	3.52	.001
	Time	0.08	0.08	1.12	.264
	Treatment:Time	0.03	0.12	0.26	.792
Confidence	Intercept	3.34	0.11	30.88	<.001
	Treatment	0.15	0.18	0.88	.383
	Time	0.06	0.05	1.32	.190
	Treatment:Time	0.09	0.08	1.22	.225
Anxiety	Intercept	3.20	0.12	27.05	<.001
	Treatment	−0.22	0.19	−1.18	.239
	Time	0.00	0.06	0.05	.961
	Treatment:Time	0.29*	0.09	3.23	.002
Effectance	Intercept	3.26	0.10	31.50	<.001
	Treatment	0.17	0.17	1.00	.317
	Time	0.01	0.05	0.28	.777
	Treatment:Time	0.30*	0.08	3.74	<.001
Teacher beliefs	Intercept	2.61	0.11	24.26	<.001
	Treatment	−0.45*	0.16	−2.80	.006
	Time	−0.05	0.06	−0.80	.423
	Treatment:Time	−0.13	0.10	−1.34	.182
Student beliefs	Intercept	3.50	0.09	39.71	<.001
	Treatment	0.18	0.13	1.41	.160
	Time	0.05	0.05	0.89	.375
	Treatment:Time	0.26*	0.08	3.04	.003

* = *p* value less than .05.

Table 5. Correlations Among Variables of Interest for *Primarily Math* Teachers and Comparison-Group Teachers.

	1	2	3	4	5	6	7	8	9
Treatment									
1. Numbers and Operations	1	.70	.63	.71	.66	.55	.15	−.44	−.01
2. Patterns, Functions, & Algebra	.70	1	.72	.60	.59	.54	.26	−.39	−.06
3. Geometry	.63	.72	1	.60	.56	.45	.08	−.47	−.01
4. Confidence	.71	.60	.60	1	.92	.77	.29	−.41	−.05
5. Anxiety	.66	.59	.56	.92	1	.76	.27	−.35	−.05
6. Effectance-Motivation	.55	.54	.45	.77	.76	1	.44	−.55	−.02
7. Student-Centered Beliefs	.15	.26	.08	.29	.27	.44	1	−.49	.04
8. Teacher-Centered Beliefs	−.44	−.39	−.47	−.41	−.35	−.55	−.49	1	−.06
Comparison Group									
1. Numbers and Operations	1	.65	.59	.43	.42	.45	.06	−.04	.02
2. Patterns, Functions, & Algebra	.65	1	.62	.57	.56	.56	.19	−.07	.05
3. Geometry	.59	.62	1	.44	.46	.49	.15	−.10	.03
4. Confidence	.43	.57	.44	1	.94	.84	.12	−.03	.12
5. Anxiety	.42	.56	.46	.94	1	.87	.17	−.02	.14
6. Effectance-Motivation	.45	.56	.49	.84	.87	1	.21	−.08	.10
7. Student-Centered Beliefs	.06	.19	.15	.12	.17	.21	1	−.38	−.04
8. Teacher-Centered Beliefs	−.04	−.07	−.10	−.03	−.02	−.08	−.38	1	.08

Table 6. Final Model Parameters for Fixed and Random Effects for the Student Impact Model (Research Questions 2 and 3).

Fixed effect	Estimate	*SE*	*t* value	*p* value	Random effect	Variance	*SD*
Intercept	6.28	0.72	8.69	<.001	Teacher	18.44	4.29
SES	0.02	0.01	0.40	.690	Teacher × Time	4.70	2.17
Participation in *Primarily Math*	1.85	0.99	1.88	.061	Residual	69.7	8.35
Fall Math Ability Score	−0.24	0.01	18.72	<.001			
Time	0.80	0.34	2.37	.021			
Numbers and Operations	−0.50	0.36	−1.39	.165			
Anxiety	1.45	0.43	3.34	.001			
Student-Centered Beliefs	−0.42	0.60	−0.70	.484			

Note. SES = socioeconomic status.

and Algebra. Neither MKT outcomes showed differential change relative to the comparison group over time. *Primarily Math* teachers started with levels of Anxiety and Effectance-Motivation similar to that of the comparison group. However, the treatment by time interaction for these attitudes toward learning outcomes were statistically significant: Anxiety was reduced 0.29 units and Effectance-Motivation increased 0.31 units per year after participating in *Primarily Math*. *Primarily Math* teachers started with statistically lower levels of Teacher-Centered Beliefs and statistically higher levels of Student-Centered Beliefs about mathematics teaching and learning. The treatment by time interaction for Student-Centered Beliefs was statistically significant: These beliefs increased 0.31 units per year of participating in *Primarily Math*.

The effects of interest for answering Research Question 1 are the treatment by time interactions and the associated t values and p values, which provide an estimate of the impact of *Primarily Math* on treatment teachers. Using a cutoff t value of 1.50 (the larger the t value the smaller the p value), interactions were statistically significantly different from zero for Numbers and Operations, Anxiety, Effectance-Motivation, and Student-Centered Beliefs. In each case, outcomes for *Primarily Math* teachers increased on average more rapidly than for comparison-group teachers.

The Impact of Primarily Math *on Student Math Ability Scores*

Table 5 shows the bivariate correlations between teacher-level predictors and student MASD averaged at the teacher level. Table 6 contains fixed and random effects for the final model. The second research question asks about the extent to which students in the classrooms of *Primarily Math* teachers differ in their mathematics achievement relative to the students of teachers in a comparison group. The main effect of treatment significantly accounted for variation in student achievement (p = .061, which meets the .10 alpha level cutoff obtained from the power analysis).

On average, *Primarily Math* teachers were estimated to have MASD change scores 1.85 units higher than teachers in

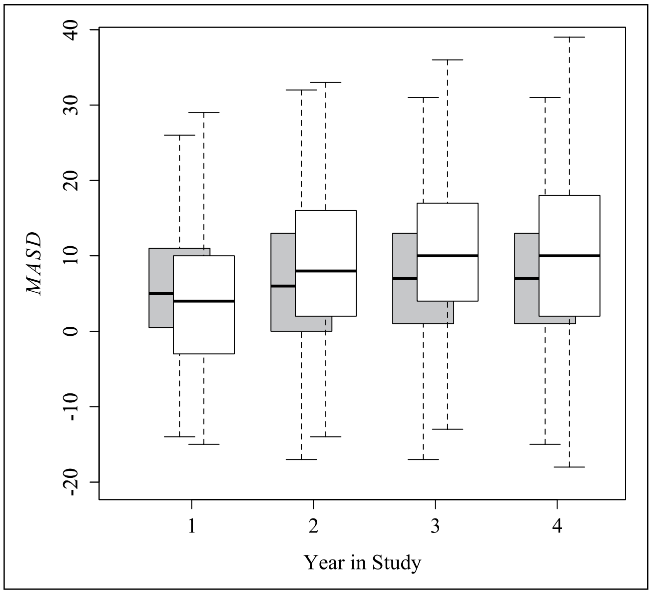

Figure 2. Observed changes in math ability scores ("MASD") by academic year ("Year in Study").

Note. Control is in gray and treatment is overlapping in white. Boxes represent the interquartile range, with the solid black horizontal lines representing the medians. Dotted lines mark the remainder of each distribution, excluding outliers. MASD = Math Ability Score Difference.

the comparison group. This effect is depicted graphically in Figure 2, which shows the observed growth in math ability scores from fall to spring (y axis) across four academic years (x axis). In Year 1, the comparison group has slightly greater MASD scores. However, from Year 2 and beyond, the treatment effect is evident in the vertical distance between the distributions for *Primarily Math* and comparison-group teachers over time.

The third research question asked what other variables were associated with student growth in math ability scores once the effect of participation (or nonparticipation) in *Primarily Math* had been accounted for. In fact, mathematics anxiety (but not teacher knowledge for teaching Numbers and Operations or Student-Centered Beliefs, t value > 1.5) still accounted for variability in growth in student math ability scores from fall to spring, regardless of participation in *Primarily Math*.

Discussion

Three groups of teachers participated in a coherent sequence of graduate-level mathematics and pedagogical coursework across 13 months. A comparison group was constructed, matched on school-level characteristics to act as the counterfactual. All district partners were located in a single state that together represent differing traditions of mathematics instruction, PD opportunities, and enrollment of students from a variety of poorly resourced and adequately resourced urban settings. This study relied upon the constructed comparison group to limit confounding school, teacher, and student differences.

Three major findings of this work warrant further discussion: the positive impact of PD on some teacher characteristics that other studies have linked to high-quality instruction and student achievement; the significant association between PD and student achievement; and the role of teachers' attitudes toward their own professional learning and its association with student achievement regardless of teacher participation in *Primarily Math*.

The Impact of Primarily Math *on Teachers*

A series of longitudinal models for all eight teacher outcomes were estimated to observe the impact (if any) of participation in our PD program on teachers. *Primarily Math* teachers' growth on four of the eight outcomes of interest were greater across time relative to teachers in a matched comparison group, in desirable directions. Specifically, *Primarily Math* teachers have greater growth in knowledge for teaching Numbers and Operations, were less anxious and more motivated to learn mathematics, and endorsed student-centered beliefs about teaching and learning.

We suggest several alternative explanations for these findings. First, self-selection could partially be at play, as on average, *Primarily Math* teachers started out differently at baseline from comparison teachers, with higher mathematical knowledge for teaching, more confident attitudes toward learning mathematics, and more progressive beliefs about teaching and student learning, and thus, perhaps, higher motivation to improve their mathematics teaching. However, while motivation was an important and possibly a distinguishing factor in who applied, motivation alone was not likely to translate into desirable teacher outcomes if there had not been a focused and coherent PD program.

Second, it is possible that teacher improvements in knowledge, attitudes, and beliefs may have been the result of state and district focus on mathematics, rather than the *Primarily Math* PD, as Nebraska was in the process of piloting its statewide mathematics achievement test in spring 2010 and implemented it in spring 2011. In fact, concurrent with *Primarily Math* coursework for Cohorts 2 and 3, districts across the state were putting greater emphasis on mathematics instruction and student achievement in mathematics. Both *Primarily Math* and comparison-group teachers participated in district-mandated PD related to mathematics (and other topics) during the study period. However, if the changes we observed in *Primarily Math* were strictly a result of district focus on mathematics, we would expect to have observed these same kinds of changes in the comparison group. Larson and Smith (2013) examined teacher results from a single district and observed that Cohort 1 seemed to have a steeper slope of mathematical knowledge for teaching gains compared with Cohort 3, particularly in 2010-2011 when the district implemented a new curriculum. At the time, Cohort 1 was participating in *Primarily Math*, but Cohort 3 was not. Larson and Smith concluded,

> It is possible that participation in year-one of *Primarily Math* allowed Cohort 1 teachers to take advantage of the associated district leadership and curriculum activities in ways that Cohort 3 teachers did not have, providing them an opportunity to integrate their knowledge and build cognitive coherence that Cohort 3 teachers did not have . . . Implementation of a new curriculum alone can have a positive effect on student achievement, but achievement gains are greater when combined with coherent and on-going professional development that emphasizes mathematics content and pedagogy. (pp. 31-32)

As we observed greater change in *Primarily Math* teacher outcomes relative to the comparison teachers, this observed growth in *Primarily Math* teachers may be attributed to their participation in our program rather than to district initiatives and activities in mathematics.

The third possible explanation is that *Primarily Math* activities benefited participating teachers. Teachers worked with mathematical *content* specific to K-3 teaching, focusing on number and operation. *Primarily Math* developers were also intentional about building a *coherent* set of learning experiences. Coursework began with mathematics immersion followed by pedagogy courses during the academic year so assignments could be tied directly to classroom practice. *Active learning* was emphasized: Mathematically rich, open-ended problems in the content courses were coupled with assignments in the pedagogy courses that required teachers to situate student thinking within learning trajectories and apply specific pedagogical techniques (e.g., Talk Moves: Chapin, O'Connor, & Anderson, 2009). *Collective participation* also was encouraged. Throughout coursework, teachers encountered challenging mathematics problems and had the opportunity to work together. Indeed, working collaboratively with fellow primary teachers fortified teachers' perseverance. The shared experience of successfully generating solutions to interesting and complex problems shifted their perceptions about their capacities to learn and engage with mathematics.

As one Primarily Math teacher reported:

> It is truly hard to pick one part or one aspect of this program. The relationships that I have made are one key aspect. As you sit through the classes you rely on one another. Maybe I really was able to grasp a concept but the person to my right is struggling. I automatically jumped in to help. The same then goes for when I was struggling. The person to my left or right or both would jump in to help. The instructors and assistants were always there to comfort and help you. I never once felt alone. I also greatly enjoyed the mathematical discussions where I was pushed to think about my own thinking.

Furthermore, the program was of sufficient *duration*: (a) teachers had time to extend their reflections, and (b) the flexibility to choose how PD could support their classroom practice. For example, one assignment required that teachers participate in a cycle of lesson planning. Teachers created or adapted a lesson, video-taped their lesson, reflected with peers on various aspects of the lesson, and then revised their lesson plan for future implementation. Engaging in these cycles of inquiry—prompted by specific questions about math teaching and learning upon which to reflect—afforded teachers the time and multiple opportunities to consolidate what they learned from their individual and collaborative experiences.

Taken together, this set of findings suggest *Primarily Math* positively impacted teacher knowledge, their attitudes toward their own professional learning, and the kinds of beliefs they endorse about mathematics teaching and learning—interrelated psychological attributes that influence pedagogical decisions.

The Impact of Primarily Math *on a Measure of Students' Mathematics Ability*

In this study, we expected the students of *Primarily Math* teachers to demonstrate greater fall-spring growth relative to the growth of students of comparison-group teachers. This expectation was partially met, as our analysis indicates from 2009 to 2013, the math ability scores of K-2 students of *Primarily Math* and comparison-group teachers both increased from fall to spring. Moreover, the effect of *Primarily Math* is greatest in the second and third year of participation (see Figure 2).

The students of comparison-group teachers also made gains beyond what would be expected. This unexpected growth may be partially or wholly accounted for by the fact that all three school districts were making major efforts to improve mathematics teaching and learning when *Primarily Math* began in 2009. Two events are worth noting. In 2010, Nebraska launched the new state accountability assessment for mathematics, launching a statewide focus on mathematics achievement. Then, in 2011, two (of three) districts who were the focus of *Primarily Math's* research agenda adopted *Math Expressions*. The philosophy and goals of this curriculum are parallel to those of *Primarily Math*. Thus, the association between *Primarily Math* and growth in students' mathematics ability may be attenuated due to district activities.

The validity of the present findings can be further supported by the quantitative and qualitative evidence, accumulated prior to this study, describing the impact of *Primarily Math* on teacher knowledge, attitudes, beliefs, and, importantly, skills related to instructional practice. Smith, Lewis, and Heaton (2013) presented evidence that students in a rural district heavily impacted by *Primarily Math* (an earlier Math–Science Partnership project) are more proficient in mathematics Grades 3 to 11 than Nebraska students in general. Smith and Shen (2012) examined artifacts and case material from nine *Primarily Math* teachers (chosen to represent low-, middle-, and high-levels of math reasoning and reflection). The researchers found changes in teachers' attitudes toward mathematics, mathematical knowledge for teaching, and mathematical teaching practices, based on their reflections and PD learning. Fleharty and Edwards (2013) also provide evidence of how participation in *Primarily Math* impacted one specific aspect of practice: the development of school-home partnerships centered on mathematics. Teachers implemented a variety of school- and home-based projects, identified and reflected upon the risks and benefits of their efforts to connect with families, and then made revisions to their projects in light of what they learned. Finally, Hopkins, Spillane, Jakopovic, and Heaton (2013) described changes in advice- and information-seeking interactions among school staff within one school system. *Primarily Math* teachers and math coaches were identified as central actors and brokers of advice and information about mathematics within and between schools. The growth in these networks was also associated with inquiry-oriented beliefs about teaching and student learning.

Teacher-Level Factors Associated With Growth in Student Mathematics Ability

In addition to examining the programmatic effect of participation in *Primarily Math*, we also tested whether any teacher-level characteristics were associated with growth in students' mathematics ability. Participation in *Primarily Math* did not moderate the relationships between teacher-level variables and growth in student mathematics ability from fall to spring.

Interestingly, there was a main effect of teacher-reported anxiety for learning mathematics: being less anxious was associated with increased student math ability scores of about 1.45 units—one of the larger effects within the model. No single individual teacher characteristic serves as the "key ingredient" for effective teaching, but we posit that to competently engage in the tasks of teaching (Ball, Hill, & Bass, 2005; Hill, Ball, & Schilling, 2008), teachers need to have a positive attitude toward learning mathematics. Teacher knowledge, attitudes, and beliefs—as previously mentioned as part of *Primarily Math's* theory of teacher change—are all interrelated psychological constructs that teachers bring to their classroom practices, but collectively and holistically impact teaching from one moment to the next. Indeed, teachers likely need to be confident and not anxious if they are to deal with the mathematics they encounter in practice, whether in cycles of planning or responding to the novel solution strategy of a child during class discussion.

Limitations

The findings of this study should be considered in light of four limitations. First, the *Primarily Math* research agenda was not designed to analyze which individual components of the PD activities were associated with teacher and student outcomes. Second, our research team recognized the importance of linking PD to changes in the quality of instructional practices. Relevant data were collected to use the Mathematical Quality of Instruction tool (MQI; LMT, 2011) to code videos of lesson plans, but these data could not be included due to range restrictions (the MQI at the time utilized a 3-point scale; the vast majority of our codes were "2"). Third, due to school district policies, we could not study the relationship of individual student characteristics (e.g., ELL status) with student outcomes. Fourth, we were not able to disaggregate findings by grade level to compare, for instance, kindergarten teachers and students with second-grade teachers and students, due to too small of a sample size of teachers, and because teachers frequently changed grade levels during the study period.

Conclusion

In the field of education, researchers often struggle to connect the impact of teacher PD to significant gains in student achievement; it is not always clear whether lack of student gains is due to a failure of the PD or other intervening factors such as the instruments not fully capturing student change. Even ideal research conditions may yield nonsignificant results (Hill, Corey, & Jacobs, 2016), very small effect sizes (Bacher-Hicks, Chin, Hill, & Staiger, 2014), or delayed results (Campbell & Malkus, 2011). Yet, in this study, our findings suggest that *Primarily Math* had a positive impact on student achievement.

Why was *Primarily Math* effective in changing teachers and affecting student outcomes? We would argue that student learning begins with teacher learning and that *Primarily Math* is well aligned to Desimone's (2009) proposed five components of effective PD. *Content* matters: Teachers cannot make sense of student math talk without having a deep understanding of the mathematics. *Duration* matters: Teachers need time to process new ideas, consolidate skills, and begin to make changes to their teaching practices. *Coherence* matters: Teachers need to see connections between PD activities and their own practice and teaching contexts. Interactions (*collective participation* and *active learning*) matter: Teachers need to be actively engaged in meaningful PD activities, and work to make sense of their learning experiences within the envelope of supportive professional communities. Competence breeds confidence; more confident teachers are better able to help students. In addition, the design of the program features summer institutes in combination with academic-year coursework that allows them to integrate new content knowledge with pedagogical knowledge and practice, as recommended by Putnam and Borko (2000).

Teacher anxiety reduction is a crucial goal for primary mathematics PD. While we expected that reducing teacher anxiety would be a positive influence on teacher and student outcomes, the size of the impact was surprising. Reducing teacher anxiety about learning mathematics was highly correlated with increased teacher confidence (Table 5). We think this confidence likely provided teachers with a platform from which to make changes to particular teaching practices, which may in turn affect how children experienced mathematics in their classrooms.

Finally, although the PD was delivered by different instructional teams over time, the National Science Foundation-funded Math–Science Partnership project had a coherent vision of effective PD, shared by project leaders (university- and school-based), instructors, and participants. This indicates that the integrity of the PD transcended particular individuals delivering the PD and represented a sustainable contribution to the culture of teaching mathematics across school districts in our state.

Future Directions

Three directions may fortify our understanding of what makes PD programs effective. First, connecting the impact of *Primarily Math* participation on the quality of instruction would provide a more complete picture of programmatic impact. Second, the *Primarily Math* study could be repeated on a larger scale, to allow for analysis of subgroups of teachers by grade level. Indeed, the content knowledge necessary to be an effective teacher for kindergarteners is different from third graders. Third, it would be instructive to further study how anxiety for learning mathematics influences how teachers benefit (or do not benefit) from PD. There is evidence that prior content knowledge partially explains the differential effect of PD on teachers, teaching, and student learning (Minor, Desimone, Lee, & Hochberg, 2016). We extend this logic to teachers who report mathematics anxiety: One size does not fit all and perhaps this subgroup of teachers need particular learning opportunities to counter their anxieties.

Appendix

How *Primarily Math* Content and Structure Align With the Core Features of Effective Professional Development.

	Primarily Math course content and structure	Examples
Content Focus	Participants took two graduate-level courses designed to strengthen conceptual knowledge of number and operation, including base-10/place value, algorithms, and rational numbers (Summer Institute 1). The third course extended into mathematics beyond Grade 3, including more rational numbers as well as geometry & algebra (Summer Institute 2).	*A particular brand of chicken nuggets come in boxes of 6, 9, and 20. What is the largest number of chicken nuggets you cannot order exactly? How do you know?* *Laura says rectangles with area 16 always have a bigger perimeter than rectangles with area 12. What would you say to Laura?*
Active Learning	Large group lecture Small group Individual work Group and individual consultation with instructor or teaching assistant upon request	Teachers video-taped themselves using two Talk Moves (Chapin, O'Connor, & Anderson, 2009), transcribed video, and completed self-reflection based on structured prompts. Brought videos and reflections to PD day and watched themselves alongside peers for debrief. Sample prompt: *"How do the Talk Moves help you to understand a child's mathematical ideas?"*
Coherence	Connections between math and math pedagogy coursework Intentional promotion of National Council of Teachers of Mathematics Practice Standards when approaching assignments In 2010, Common Core State Standards rolled out and its content integrated into coursework	The Child Study was a long-term class project that required teachers to observe a target child over 8-10 weeks and situate their learning within Clements and Sarama's (2009) trajectories. Unit planning assignment where teachers used local curriculum.
Duration	Summer Institutes 2 weeks long, each 8 hr days Long-term projects, with built-in opportunities for reflection and revision take place over the course of the semester (For a more in-depth example, see Fleharty & Edwards, 2013)	A minimum of 80 contact hours during each summer institute. Homework each night, where instructor and teaching assistants were available to support individual and group efforts at completing homework. Long-term projects assigned over course of school year. Teachers received instructor and peer feedback at midpoint to deadline.
Collective Participation	In-class assignments facilitated collaboration between teachers who taught at same or different grade levels Cohorts built around teachers working in same school district Multiple teachers at same school (different grades) participated	During academic-year courses, teachers provided feedback to peers on various assignments (e.g., professional writings) before submitting to instructors. In math courses, teachers worked almost exclusively in cooperative groups to solve problems and make sense of multiple-solution paths.

Note. PD = professional development.

Authors' Note

Since the completion of this research, affiliation information for two of the authors has changed. Lixin Ren's affiliation is now East China Normal University. W. James Lewis affiliation now adds the National Science Foundation (temporary appointment).

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Notes

1. More information on *Primarily Math* coursework can be found at http://scimath.unl.edu/csmce/course-materials/
2. *Primarily Math* teachers reported three reasons for stopping *Primarily Math* participation: (a) moved out of state or retired, (b) assigned to a position outside of K-3, or (c) personal reasons (e.g., illness; pregnancy).
3. In the first power analysis, achieving 80% power at the alpha level .05, given the maximum number of teachers *Primarily Math* could handle per cohort, was unattainable.
4. Further details of our factor analyses and results are presented in Ren and Smith (2013).
5. Analyses of teacher data are explored in far more depth in Kutaka et al. (2016).
6. Further details about the teacher and student model specification (including notation) can be obtained by contacting the first author.

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